Er Diagram

Metadata

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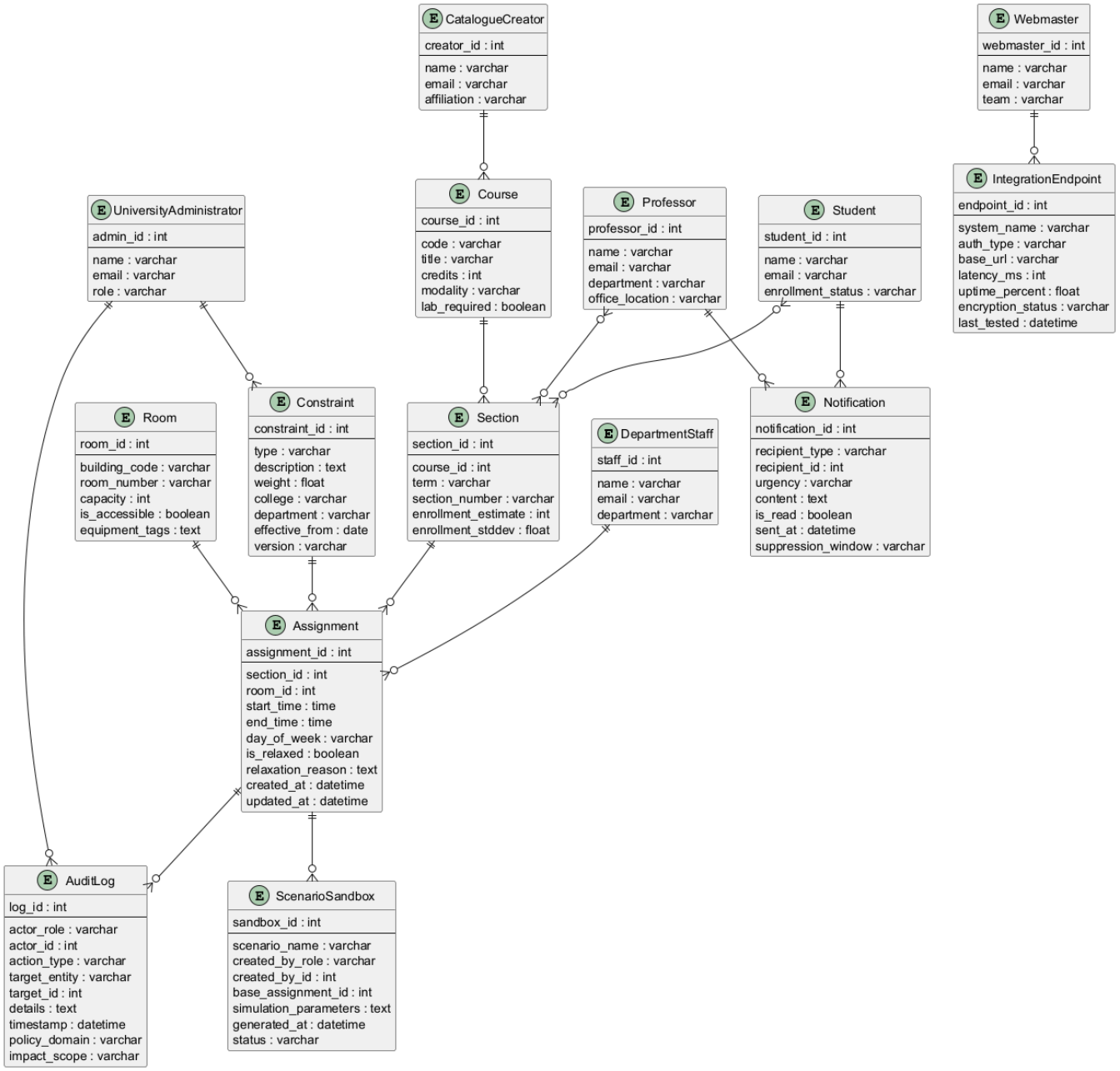
Diagrams

Diagram 1: Diagram Base64



Entities

Name	Attributes
DepartmentStaff	['staff_id : int', '--', 'name : varchar', 'email : varchar', 'department : varchar']
UniversityAdministrator	['admin_id : int', '--', 'name : varchar', 'email : varchar', 'role : varchar']
CatalogueCreator	['creator_id : int', '--', 'name : varchar', 'email : varchar', 'affiliation : varchar']
Webmaster	['webmaster_id : int', '--', 'name : varchar', 'email : varchar', 'team : varchar']
Student	['student_id : int', '--', 'name : varchar', 'email : varchar', 'enrollment_status : varchar']
Professor	['professor_id : int', '--', 'name : varchar', 'email : varchar', 'department : varchar', 'office_location : varchar']
Course	['course_id : int', '--', 'code : varchar', 'title : varchar', 'credits : int', 'modality : varchar', 'lab_required : boolean']
Section	['section_id : int', '--', 'course_id : int', 'term : varchar', 'section_number : varchar', 'enrollment_estimate : int', 'enrollment_stddev : float']
Room	['room_id : int', '--', 'building_code : varchar', 'room_number : varchar', 'capacity : int', 'is_accessible : boolean', 'equipment_tags : text']
Assignment	['assignment_id : int', '--', 'section_id : int', 'room_id : int', 'start_time : time', 'end_time : time', 'day_of_week : varchar', 'is_relaxed : boolean', 'relaxation_reason : text', 'created_at : datetime', 'updated_at : datetime']
Constraint	['constraint_id : int', '--', 'type : varchar', 'description : text', 'weight : float', 'college : varchar', 'department : varchar', 'effective_from : date', 'version : varchar']
AuditLog	['log_id : int', '--', 'actor_role : varchar', 'actor_id : int', 'action_type : varchar', 'target_entity : varchar', 'target_id : int', 'details : text', 'timestamp : datetime', 'policy_domain : varchar', 'impact_scope : varchar']
Notification	['notification_id : int', '--', 'recipient_type : varchar', 'recipient_id : int', 'urgency : varchar', 'content : text', 'is_read : boolean', 'sent_at : datetime', 'suppression_window : varchar']
ScenarioSandbox	['sandbox_id : int', '--', 'scenario_name : varchar', 'created_by_role : varchar', 'created_by_id : int', 'base_assignment_id : int', 'simulation_parameters : text', 'generated_at : datetime', 'status : varchar']
IntegrationEndpoint	['endpoint_id : int', '--', 'system_name : varchar', 'auth_type : varchar', 'base_url : varchar', 'latency_ms : int', 'uptime_percent : float', 'encryption_status : varchar', 'last_tested : datetime']

Relationships

From Entity	To Entity	Relationship Type	Label
DepartmentStaff	Assignment	one-to-many	reviews_and_approves
UniversityAdministrator	Constraint	one-to-many	configures
UniversityAdministrator	AuditLog	one-to-many	generates
CatalogueCreator	Course	one-to-many	manages
Webmaster	IntegrationEndpoint	one-to-many	configures

From Entity	To Entity	Relationship Type	Label
Student	Notification	one-to-many	receives
Professor	Notification	one-to-many	receives
Course	Section	one-to-many	has_sections
Room	Assignment	one-to-many	hosts
Section	Assignment	one-to-many	assigned_to
Assignment	AuditLog	one-to-many	triggers
Assignment	ScenarioSandbox	one-to-many	used_in_simulation
Constraint	Assignment	one-to-many	applied_to
Student	Section	many-to-many	enrolled_in
Professor	Section	many-to-many	teaches

Uml Fmt Content

@startuml
entity DepartmentStaff {
staff_id : int

name : varchar
email : varchar
department : varchar
}

entity UniversityAdministrator {
admin_id : int

name : varchar
email : varchar
role : varchar
}

entity CatalogueCreator {
creator_id : int

name : varchar
email : varchar
affiliation : varchar
}

entity Webmaster {
webmaster_id : int

name : varchar
email : varchar
team : varchar
}

entity Student {
student_id : int

name : varchar
email : varchar
enrollment_status : varchar
}

entity Professor {
professor_id : int

name : varchar
email : varchar
department : varchar
office_location : varchar
}

entity Course {
course_id : int

code : varchar
title : varchar
credits : int
modality : varchar
lab_required : boolean
}

entity Section {
section_id : int

course_id : int
term : varchar
section_number : varchar
enrollment_estimate : int
enrollment_stddev : float
}

entity Room {
room_id : int

building_code : varchar
room_number : varchar
capacity : int
is_accessible : boolean
equipment_tags : text
}

entity Assignment {
assignment_id : int

section_id : int
room_id : int
start_time : time
end_time : time
day_of_week : varchar
is_relaxed : boolean
relaxation_reason : text
created_at : datetime
updated_at : datetime
}

entity Constraint {
constraint_id : int

type : varchar
description : text
weight : float
college : varchar
department : varchar
effective_from : date
version : varchar
}

entity AuditLog {
log_id : int

actor_role : varchar
actor_id : int
action_type : varchar
target_entity : varchar
target_id : int
details : text
timestamp : datetime
policy_domain : varchar
impact_scope : varchar
}

entity Notification {
notification_id : int

recipient_type : varchar
recipient_id : int
urgency : varchar
content : text
is_read : boolean
sent_at : datetime
suppression_window : varchar
}

entity ScenarioSandbox {
sandbox_id : int

scenario_name : varchar
created_by_role : varchar
created_by_id : int
base_assignment_id : int
simulation_parameters : text
generated_at : datetime
status : varchar
}

entity IntegrationEndpoint {
endpoint_id : int

system_name : varchar
auth_type : varchar
base_url : varchar
latency_ms : int
uptime_percent : float
encryption_status : varchar
last_tested : datetime
}

DepartmentStaff ||--o{ Assignment
UniversityAdministrator ||--o{ Constraint
UniversityAdministrator ||--o{ AuditLog
CatalogueCreator ||--o{ Course
Webmaster ||--o{ IntegrationEndpoint
Student ||--o{ Notification
Professor ||--o{ Notification
Course ||--o{ Section
Room ||--o{ Assignment
Section ||--o{ Assignment
Assignment ||--o{ AuditLog
Assignment ||--o{ ScenarioSandbox
Constraint ||--o{ Assignment
Student }o--o{ Section
Professor }o--o{ Section
@enduml

Summary

The ER diagram models core entities of the University Class Scheduler (UCS), capturing roles, academic resources, scheduling assignments, constraints, audit logs, notifications, simulations, and integrations, with relationships reflecting real-world operational dependencies and compliance requirements.